

Milestone 3 Summary Document

Cannon Blocker

CS 4412 Data Mining

Implementation Status:

The two algorithms that I have implemented are the K-means clustering and the Apriori algorithm. Both of these algorithms have worked great to find and support evidence for my two discovery questions. Both of these algorithms have shown unique ways in which beatmaps are different from each other and what actually defines the distinct groups of songs found in the clustering and further supported by the apriori algorithm.

Explored Approaches:

I first explored the K-means clustering for Milestone 2 after prepping all of the data, and this algorithm turned out to be a success and was able to split the beatmaps into 4 distinct clusters, each with their own unique attributes. It showed that there were distinct groups of maps, even if they may overlap a bit. It also showed how stats like BPM and difficulty rating are very important in determining the popularity of a map and its favorite count, as was originally found in the heatmap from the EDA section.

I also explored the DBSCAN as I wanted to see if it could show me any new results or insights not found in the K-means clustering. However, the DBSCAN did not work as intended and despite finding 6 unique clusters, cluster 0 was ~5600 beat maps and cluster 1 was about ~55000 beat maps, with clusters 2-6 being only like 5 maps each. After looking more deeply at what happened, it turns out that when the data is very closely related and a lot of metrics are similar across hundreds of beat maps, the DBSCAN tends to just group it all together and not really provide any useful insights. This was also a problem in the last milestone and was why I had to force the K-means to 4 clusters instead of the 2 clusters that the silhouette score recommended because it would just group all of the beat maps into one large cluster.

Finally, I explored the Apriori algorithm, which I originally thought wouldn't work very well because I thought it would not fit my dataset as each map is a distinct object, unrelated to the others. However, thinking of the attributes as the objects to use in the Apriori turned out to be a good approach, and I was able to find some good insights to support my discovery questions.

Results and Discovery Question Progress:

First Mining Results:

The k-means graph with k=4 showed that there were distinct groups of maps, even if they may overlap a bit. It also showed how stats like BPM and difficulty rating are very important in determining the popularity of a map and its favorite count, as was originally found in the heatmap from the EDA section.

How the results relate to the discovery questions:

Question 1: What map and song characteristics are most commonly shared among the most popular beatmaps?

By clustering, we can see that "popular" maps (high favorite and decent plays) do not all look the same. Cluster 2 reveals a highly popular segment driven by high BPM (~180+) and a high difficulty (~5). In contrast cluster 0 achieves popularity through shorter, easier gameplay (primarily targeting newer players).

Question 2: Are there distinct groups of songs that achieve popularity in different ways (high replay vs high appreciation)?

Cluster 2 (The Grind), perfectly defines the high replay category. The replay ratio (2.25) is massively higher than any other cluster. These maps have high difficulty, but they still achieve lots of plays from players constantly failing and retrying them, inflating their playcount relative to their passcount.

Cluster 0 (Experiences), represents the maps within the high appreciation category of maps. They have very low replay ratios (meaning players often pass them on the first few tries) but maintain incredibly high favorite counts. Players play them for the music, the storyboard, or the mapping quality, not to grind performance points or difficulty.

Second Mining Results:

The Apriori algorithm with minimum support 0.4 further confirmed the findings from the K-means mining results. It mathematically showed that difficulty and standard BPM defined as 150-200 BPM are the driving factors in the popularity of a map, specifically its play count. It also showed that through the absence of a link between High Replay and High Appreciation, there are distinct groups of beat maps.

Connections to The Discovery Questions

Question 1: What map and song characteristics are most commonly shared among the most popular beatmaps?

Using the apriori algorithm, we can see that the primary driver for popular (player retention and high replay ability) maps is difficulty. Rules 1, 2 and 3, the strongest rules, all revolve around difficulty.

Rule 3 shows that if a map is Expert (>5.5*), there is a massive 90% confidence that it will result in High Replay, with a lift of 3.60. This demonstrates a highly robust statistical association that extreme difficulty is the single most consistent characteristic shared by highly engaged maps.

Rule 2 refines this finding further. When you combine Expert (>5.5*) difficulty with Standard BPM (150-200), the confidence of hitting High Replay jumps to 91% with a lift of 3.63. Even for slightly easier maps (Rule 8), the Standard (2m-4m) length and Standard BPM still strongly link to High Replay.

The most engaging maps in osu! share a strict mechanical blueprint: they are highly difficult (3.5 stars and above) and adhere to standard rhythm game tempos (150-200 BPM).

Question 2: Are there distinct groups of songs that achieve popularity in different ways (high replay vs high appreciation)?

Looking at rules 4, 6, 10, 11, 13, and 14, in all of these rules, High Replay is explicitly pairs with Normal Appreciation. The algorithm proves that players aggressively replaying a Hard/Insane map does not translate into them favoring it. They are playing for mechanical mastery or performance points (PP), not because they love the song or the artistry of the map.

In the top 15 rules listed above, there is not a single rule that links High replay to High Appreciation. If these two metrics were just different flavors of "good maps", they would appear together. Their complete separation in the highest-lift rules mathematically validates the findings from the K-Means clustering.

The apriori algorithm statistically proves that Osu! maps achieve popularity through mutually exclusive avenues. "The Grind" is driven by difficulty, standard BPM, and generates massive playcounts without many favorites. "The Experience" (High Appreciation) exists entirely outside of the high-difficulty, high-retry loop.

M4 Plan:

For polish and presentation, I want to look a little deeper at the results and see if I can find any new discoveries that I haven't yet found. While the discussion questions are sufficiently answered, I want to find other insights from the data and algorithms.